

Stepper Motor Formulae

DM542E Microstep Driver — Standard Math Notation

1/16 Microstepping | PPR = 3200 | Step Angle = 0.1125°

DM542E — SWITCH CONFIGURATION

Highlighted row = configuration used in all examples.

SW5	SW6	SW7	SW8	Microstep	PPR
ON	ON	ON	ON	1/16	3200
ON	ON	OFF	ON	1/8	1600
ON	ON	ON	OFF	1/4	800
ON	ON	OFF	OFF	1/2	400
OFF	OFF	OFF	OFF	Full	200

Example baseline (all formulae): PPR = 3200 | Step Angle = 0.1125° | RPM = 300 | Frequency = 16000 Hz | Rotations = 5 | Speed = 8000 Hz

FORMULAE WITH STEP-BY-STEP DERIVATION (STANDARD MATH / FRACTION NOTATION)

1. Step Angle from PPR

Finds angular distance per microstep pulse.

$$\text{Step Angle} = \frac{360}{\text{PPR}}$$

Given: PPR = 3200 (DM542E at 1/16 microstep)

Step 1: Substitute PPR = 3200 into the formula

$$\text{Step Angle} = \frac{360}{3200}$$

Step 2: Divide

✓ Step Angle = 0.1125° per pulse

2. PPR from Step Angle

Reverse-calculates PPR when step angle is known.

$$\text{PPR} = \frac{360}{\text{Step Angle}}$$

Given: Step Angle = 0.1125°

Step 1: Substitute Step Angle = 0.1125°

$$\text{PPR} = \frac{360}{0.1125}$$

Step 2: Divide

✓ PPR = 3200 pulses / revolution

3. PPR from Frequency & RPM

Derives PPR from measured pulse frequency and shaft speed.

$$\text{PPR} = \frac{\text{Frequency} \times 60}{\text{RPM}}$$

Given: Frequency = 16000 Hz | RPM = 300

Step 1: Substitute values

$$\text{PPR} = \frac{16000 \times 60}{300}$$

Step 2: Simplify numerator: 16000 × 60 = 960000

$$\text{PPR} = \frac{960000}{300}$$

Step 3: Divide

✓ PPR = 3200 pulses / revolution

4. RPM from Frequency & PPR

Calculates shaft speed in RPM from pulse frequency.

$$\text{RPM} = \frac{\text{Frequency} \times 60}{\text{PPR}}$$

Given: Frequency = 16000 Hz | PPR = 3200

Step 1: Substitute values

$$\text{RPM} = \frac{16000 \times 60}{3200}$$

Step 2: Simplify numerator: $16000 \times 60 = 960000$

$$\text{RPM} = \frac{960000}{3200}$$

Step 3: Divide

✓ **RPM = 300 rev / minute**

5. Pulse Frequency

Required step pulse frequency to achieve a target RPM.

$$\text{Frequency} = \frac{\text{PPR} \times \text{RPM}}{60}$$

Given: PPR = 3200 | RPM = 300

Step 1: Substitute values

$$\text{Frequency} = \frac{3200 \times 300}{60}$$

Step 2: Simplify numerator: $3200 \times 300 = 960000$

$$\text{Frequency} = \frac{960000}{60}$$

Step 3: Divide

✓ **Pulse Frequency = 16000 Hz (16 kHz)**

6. Pulses per Step

Number of pulses to move shaft by one full step angle.

$$\text{Pulses/Step} = \frac{\text{PPR} \times \text{Step Angle}}{360}$$

Given: PPR = 3200 | Step Angle = 0.1125°

Step 1: Substitute values

$$\text{Pulses/Step} = \frac{3200 \times 0.1125}{360}$$

Step 2: Simplify numerator: $3200 \times 0.1125 = 360$

$$\text{Pulses/Step} = \frac{360}{360}$$

Step 3: Divide

✓ Pulses/Step = 1 pulse per microstep (0.1125°)

7. Total Pulses for Multiple Rotations

Total pulses to complete a set number of full revolutions.

$$\text{Total Pulses} = \text{Pulses/Step} \times \text{PPR} \times \text{Rotations}$$

Given: Pulses/Step = 1 | PPR = 3200 | Rotations = 5

Step 1: Substitute values

$$\text{Total Pulses} = 1 \times 3200 \times 5$$

Step 2: Multiply

✓ Total Pulses = 16,000 pulses for 5 rotations

8. Estimated Time

Duration to deliver all required pulses at a given step rate.

$$\text{Time} = \frac{\text{Total Pulses}}{\text{Speed (Hz)}}$$

Given: Total Pulses = 16000 | Speed = 8000 Hz

Step 1: Substitute values

$$\text{Time} = \frac{16000}{8000}$$

Step 2: Divide

✓ Time = 2.0 seconds

9. Speed from Pulses & Time

Required step frequency to complete motion within a time limit.

$$\text{Speed} = \frac{\text{Total Pulses}}{\text{Time (s)}}$$

Given: Total Pulses = 16000 | Time = 2 s

Step 1: Substitute values

$$\text{Speed} = \frac{16000}{2}$$

Step 2: Divide

✓ Speed = 8000 Hz (8 kHz)

10. RPM from Speed & PPR

Converts step pulse frequency back to shaft speed in RPM.

$$\text{RPM} = \frac{\text{Speed (Hz)} \times 60}{\text{PPR}}$$

Given: Speed = 8000 Hz | PPR = 3200

Step 1: Substitute values

$$\text{RPM} = \frac{8000 \times 60}{3200}$$

Step 2: Simplify numerator: $8000 \times 60 = 480000$

$$\text{RPM} = \frac{480000}{3200}$$

Step 3: Divide

✓ RPM = 150 rev / minute

QUICK REFERENCE SUMMARY

#	Formula (fraction notation)	Solves for	Result
1	$360 / \text{PPR}$	Step Angle	0.1125°
2	$360 / \text{Step Angle}$	PPR	3200 PPR
3	$(f \times 60) / \text{RPM}$	PPR	3200 PPR
4	$(f \times 60) / \text{PPR}$	RPM	300 RPM
5	$(\text{PPR} \times \text{RPM}) / 60$	Frequency	16000 Hz
6	$(\text{PPR} \times \text{Step Angle}) / 360$	Pulses/Step	1 pulse
7	$\text{Pulses/Step} \times \text{PPR} \times \text{Rotations}$	Total Pulses	16,000
8	$\text{Total Pulses} / \text{Speed}$	Time	2.0 s
9	$\text{Total Pulses} / \text{Time}$	Speed	8000 Hz
10	$(\text{Speed} \times 60) / \text{PPR}$	RPM	150 RPM

Stepper Control App — Step Angle Page

The app's **Calculated Values** panel uses three formulae (6, 7, 8) from this document. The table below shows what the app computes from the inputs visible on screen, followed by corrected values for proper DM542E operation at 300 RPM.

Input Values Comparison

Field	App Screen Value	Field	Correct for 300 RPM
PPR	3200	PPR	3200
Microstep	16×	Microstep	16×
Step Angle	0.1125°	Step Angle	0.1125°
Rotations	5	Rotations	5
Speed (Hz)	1000 ■	Speed (Hz)	16000 ✓

App Calc 1 — Pulses / Step (Formula 6)

How many controller pulses equal one microstep movement.

$$\text{Pulses/Step} = \frac{\text{PPR} \times \text{Step Angle}}{360}$$

Given: PPR = 3200 | Step Angle = 0.1125°

Step 1: Substitute values

$$\text{Pulses/Step} = \frac{3200 \times 0.1125}{360}$$

Step 2: Simplify numerator: $3200 \times 0.1125 = 360$

$$\text{Pulses/Step} = \frac{360}{360}$$

Step 3: Divide

✓ Pulses/Step = 1 (app displays: 1 ✓ Correct)

App Calc 2 — Total Pulses (Formula 7)

Total pulses to complete all rotations.

Total Pulses = Pulses/Step x PPR x Rotations

Given: Pulses/Step = 1 | PPR = 3200 | Rotations = 5

Step 1: Substitute values

Total Pulses = 1 x 3200 x 5

Step 2: Multiply

✓ **Total Pulses = 16,000 pulses for 5 rotations**

■ **App displays: 5** — Because app uses Step Angle (0.1125°) as rotations input misread, not the full PPR x Rotations chain. Correct value = 16,000.

App Calc 3 — Estimated Time (Formula 8)

Time to complete all pulses at the set speed.

Time = $\frac{\text{Total Pulses}}{\text{Speed (Hz)}}$

Scenario A — As shown in app (Speed = 1000 Hz, Total Pulses = 5)

Given: Total Pulses = 5 (app value) | Speed = 1000 Hz (app value)

Step 1: Substitute

Time = $\frac{5}{1000}$

Step 2: Divide

✓ **Time = 0.005 s ≈ 0.01 s (app displays: 0.01 s ✓ matches)**

Scenario B — Correct values (Speed = 16000 Hz, Total Pulses = 16000)

Given: Total Pulses = 16,000 (correct) | Speed = 16000 Hz (correct for 300 RPM)

Step 1: Substitute

Time = $\frac{16000}{16000}$

Step 2: Divide

✓ **Time = 1.0 second for 5 full rotations at 300 RPM**

APP CALCULATED VALUES — SUMMARY COMPARISON

App Field	Formula	App Value (1000 Hz)	Correct Value (16000 Hz)
Pulses / Step	Formula 6: (PPR x Step°) / 360	1	1
Total Pulses	Formula 7: Pulses/Step x PPR x N	5 ■	16,000 ✓
Est. Time	Formula 8: Total Pulses / Speed	0.01 s ■	1.0 s ✓

